From TriBE to GemmaBE

Abstract— **Brain encoding systems like TriBE v2 achieve state-of-the-art performance by combining three specialized backbones (V-JEPA2 for video, Wav2Vec-BERT for audio, LLaMA 3.2 for text) with late fusion. This raises a fundamental question: can a single frozen multimodal LLM replace this complex pipeline? We present GemmaBE, which uses Gemma 4 E2B-it (5.1B parameters, frozen) as a unified feature extractor for video, audio, and text, producing a single 1536-dimensional embedding per TR through early fusion. Unlike TriBE's late fusion, where modalities remain separate until the final layers, Gemma 4 projects all inputs into a shared language space from the earliest layers, implicitly learning cross-modal interactions during pre-training. We evaluate this hypothesis on the Algonauts 2025 (NeuroMod) dataset (~50 hours of naturalistic stimuli, 2 subjects), predicting 1000 cortical parcels (Schaefer-1000). Through systematic ablations, we test: (1) whether temporal dependencies in the embeddings benefit from a Transformer (4 layers, 8 heads, 67-TR windows) versus a pointwise MLP; (2) whether visual and auditory modalities contribute beyond text-only input; and (3) whether the model generalizes to never-seen content via strict episodic hold-out (Season 6 as test). Results demonstrate that Gemma 4's high-level narrative embeddings reach ~0.10 Pearson correlation on held-out episodes, comparable to a linear Ridge baseline, but without benefiting from deep temporal architectures or generalizing through early multimodal fusion. These findings indicate that generalist LLMs capture linearly decodable semantic information but lack the low-level sensory representations necessary for robust cross-episode brain encoding.**

Keywords—Brain Encoder, Multimodal, fRMI, Late fusion, Temporal Transformer, stimulus, embeddings, Pearson

I. INTRODUCTION

Brain encoding predicts brain activity (fMRI) from external stimuli, seeking to understand what representations the brain learns and to develop AI more aligned with human cognition [1]. Traditionally, linear models such as Ridge Regression have been used, but recent advances with DNNs and Transformers have significantly improved accuracy [2].

A recent milestone is TriBE v1 and v2 (Meta), the Algonauts 2025 winner that achieves high-resolution predictions using three specialized foundational models (Wav2Vec-BERT for audio, V-JEPA2 for video, LLaMA 3.2 for text) whose representations are concatenated and processed with a Transformer Encoder [3], [4]. However, this architecture presents limitations: massive hardware requirements for extraction, and a late fusion of modalities that does not exploit cross-interactions in early layers [5].

In this work, we propose GemmaBE, an architectural simplification that replaces the decoupled trimodal backbone of TriBE v1 and v2 with a single frozen multimodal model (Gemma 4 E2B-it). With the objective of evaluating its viability as a feature extractor for brain encoding, determining: (1) whether its early-fusion embeddings generalize to unseen content through strict episodic hold-out; (2) what visual and auditory modalities contribute compared to text alone; and (3) whether temporal dependencies in the embeddings benefit from a Transformer over a pointwise MLP. The experiments are run on Algonauts 2025 (NeuroMod, ~50h, 2 subjects), predicting 1000 cortical parcels (Schaefer-1000).

II. METHODOLOGY

The most complex part of brain decoding is obtaining a high-quality dataset of a significant size for neural networks to learn from. To predict brain activity, naturalistic stimuli are needed, such as auditory, visual, and textual ones. The most common way to obtain these brain signals is through a magnetic resonance imaging (MRI) machine [6].

The NeuroMod team has made a contribution to the neuroscientific community and for years has built a dataset with more than 200 hours of naturalistic stimuli from 5 different subjects [7]; for the development of this model, approximately 50 hours of visual, auditory, and textual stimuli from the sitcom *Friends* were used, spanning from season 1 to 6, which are separated into 12-minute chunks to promote subject rest, and additionally, there are 10 hours from 5 featured movies [8].

The data collected by the MRI machine have a repetition time (TR) of 1.49 seconds [6]. Each captured brain volume is normalized and aligned to a standard spatial template, known as the MNI (Montreal Neurological Institute) space [6]. Subsequently, a mathematical model known as the Schaefer atlas is applied [8], [9], which allows collapsing thousands of three-dimensional pixels (voxels) into clusters of 1,000 functionally defined parcels to obtain a single average time series per parcel [8], [9]. This acts as a dimensionality reduction tool that drastically decreases the computational cost necessary for model training, by avoiding the individual mapping of thousands of voxels (original resolution), many of which are not representative or provide redundant information [3], [9].

One of the most important phases is stimulus extraction. This work uses Gemma 4 E2B-it (5.1B parameters) [10], an open-source multimodal model with a context window of ~128k tokens and the ability to process video, audio, and text in a single forward pass [10]. For each TR (1.49s) [6], the model receives: (1) 32 equally spaced frames from the last 32 seconds of video; (2) 30 seconds of previous audio at 16kHz; and (3) the last 1,024 words (1,300 tokens after tokenization). Following the Gemma team's recommendations, images and audio are placed at the beginning of the prompt, followed by text [10]. Unlike TriBE, where audio is processed bidirectionally (Wav2Vec-BERT) [3], Gemma 4 operates causally, processing only past information.

Gemma 4 E2B consists of 35 attention layers [11]. For extraction, selective hooks are registered on layers 20, 25, 30, and 35, chosen for their depth and periodic distribution, combined with output_hidden_states=False to minimize memory usage. Subsequently, the activations of the 4 layers are averaged by blocks (layers 20-25 and 30-35), normalized, and a temporal pooling with padding masking is applied, resulting in a single embedding of 1,536 dimensions per TR [11].

For feature extraction, the Meta team (in the TriBE v1 and v2 architectures) constructs a uniformly spaced grid at a frequency of 2Hz [4]. The purpose of this high temporal resolution (every 0.5 seconds) is to precisely align the text features with those of the audio and video before processing [4].

Subsequently, the output of the temporal model passes through an adaptive average pooling layer that compresses the sequence, decimating the stimulus frequency to generate a single embedding aligned to the fMRI acquisition rate. In our case, to optimize computational cost and maintain synchrony with the BOLD response, temporal stimulus extraction was processed directly at a frequency of 1Hz.

Approximately 51 hours of stimuli were extracted in two conditions: multimodal (video+audio+text) and text-only, generating ~123,000 TRs per subject and condition, with a hemodynamic alignment of 5 seconds to compensate for the delay between stimulus presentation and the peak of the BOLD response [12].

The model architecture, illustrated in Fig. 1, is inspired by TriBE v1 and v2 [3], [4]. Initially, there is a Bottleneck that compresses the output of Gemma 4 (1,536 dimensions) into a compact representation of 512 dimensions through a linear projection. The core component is a Temporal Transformer with 4 layers and 8 attention heads, which processes windows of 67 consecutive TRs (~100 seconds). This temporal window is fundamental because it allows the model to capture long-range sequential dependencies in naturalistic stimuli, like TriBE [3].

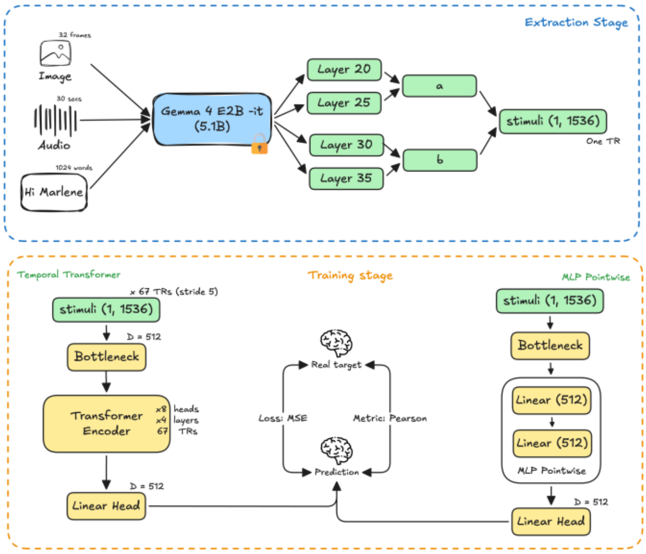

Fig. 1. Architecture of GemmaBE. (A) The frozen multimodal backbone Gemma 4 E2B-it (5.1B parameters) extracts 1,536-dimensional embeddings per TR from video (32 frames), audio (30s), and text (1,024 words). (B) The Bottleneck reduces dimensionality to 512. (C) The Temporal Transformer processes 67-TR windows (~100s) with 8 self-attention layers. (D) The linear Head projects to 1,000 parcels (Schaefer-1000). The ablation replaces the Transformer with a pointwise MLP with 2 hidden layers (dotted line).

The output of the Transformer is projected through a final linear layer (head) to the 1,000 parcels of the Schaefer-1000 atlas, matching the dimensionality of the target fMRI signal [9]. As an architectural ablation, an alternative model composed of a Pointwise MLP with 2 hidden layers of 512 dimensions was built, completely omitting the temporal attention mechanism. This comparison allows for isolating the effect of the full Transformer (~1.8M trainable parameters).

Maintaining the temporal alignment between stimulus and brain response is fundamental to preserving dataset coherence [12]. The hemodynamic response introduces a natural delay of approximately 5 seconds between stimulus presentation and the registered fMRI activation peak [12]. Therefore, to predict the BOLD activity at time T, the corresponding stimulus at time T- Δ is used, where Δ = 5 seconds (equivalent to 3 TRs given our TR of 1.49 seconds):

$$stimulus[T - \Delta] \rightarrow BOLD[T] \quad (1)$$

The training hyperparameters are based on TriBE v1 [3] with adaptations for our hardware. The AdamW optimizer is used with a learning rate of 10^{-4}, weight decay of 10^{-5}, and a Cosine Annealing scheduler over a maximum of 100 epochs. The training employs 16-bit mixed precision and stops early if the Pearson correlation in validation does not improve for 20 consecutive epochs. The dataset is divided into a training set of seasons 1-5 of Friends and 5 movies (~100k TRs); a test set of the full season 6 of Friends (~23k TRs), never seen during training; and a validation set of a random 10% from TRAIN, used for early stopping. The strict test set avoids data leakage by overlapping temporal windows (stride 5, window=67).

The standard brain encoding metric is the Pearson correlation [2], [13] per parcel, between the predicted and real fMRI signals. During training, the mean squared error (MSE) is minimized, but the performance is reported in Pearson, since it measures the linear correspondence between neural time series [8]. For each of the 1,000 vertices of the Schaefer-1000 atlas [9], Pearson was computed between the prediction and the real BOLD to report the average Pearson and standard deviation across all parcels; the percentage of parcels with r>0.15, a threshold we consider significant to demonstrate the viability of the architecture, and the test MSE as a secondary metric.

III. Results

In Table I we present the performance on the test set, consisting of the 49 chunks of Friends Season 6 (~22,949 TRs, ~9.5 hours of content never seen during training). This is the only number that reflects real generalization.

TABLE I. Mean Pearson test per parcel (Season 6 hold-out)

Model	Stimulus	Sub-01	Sub-02	Media
Transformer (with HRF)	Multimodal	0.067	0.074	0.071
Transformer (with HRF)	Text-only	0.069	0.079	0.072
Linear (MLP)	Multimodal	0.073	0.079	0.076
Linear (MLP)	Text-onluy	0.074	0.078	0.076
Ridge (linear biseline)	-	0.100	0.111	0.105

Table II shows the validation results (random 10% of S1-S5). These values are inflated because the 67-TR windows with stride = 5 overlap between train and val (93% overlap), allowing the memorization of specific scenes.

TABLE II. Validation Pearson (S1-S5)

Model	Stimulus	Sub-01	Sub-02	Media
Transformer (with HRF)	Multimodal	0.752	0.745	0.748
Transformer (with HRF)	Text-only	0.751	0.744	0.748
Linear (MLP)	Multimodal	0.357	0.329	0.343
Linear (MLP)	Text-only	0.357	0.328	0.343

The Transformer model reduces its performance from ~0.75 (val) to ~0.07 (test), a drop of ~0.68. The Linear model drops from ~0.34 to ~0.08 (drop of ~0.26). The difference in the drop suggests that the Transformer memorizes the S1-S5 scenes more efficiently, but both models fail equally on S6.

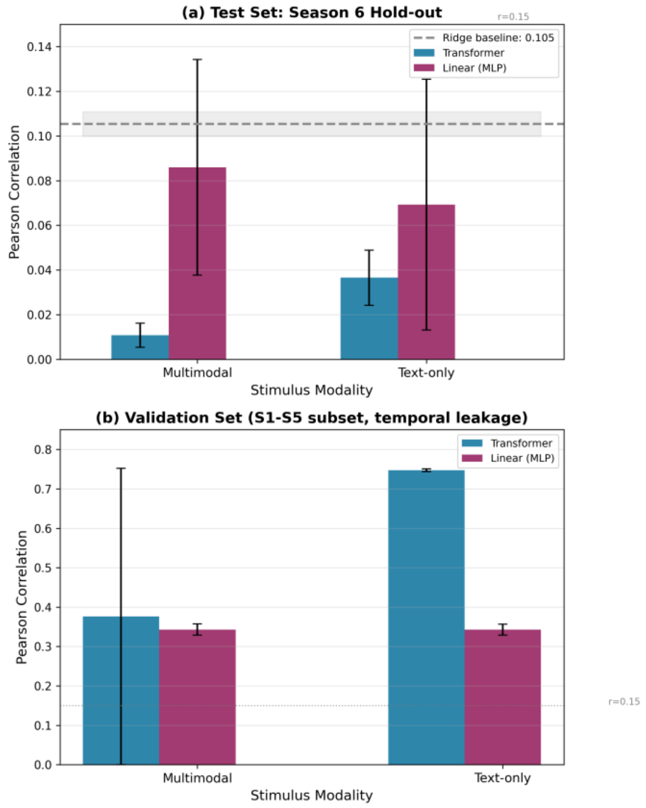

Fig. 2. Brain encoding performance by architecture and modality. Panel (a): Test set (Season 6 hold-out). All models achieve ~0.07-0.11 Pearson. Ridge baseline (dotted line) slightly outperforms the neural models. Panel (b): Validation set (S1-S5 with overlapping windows). The artificial gap between Transformer and Linear is evident.

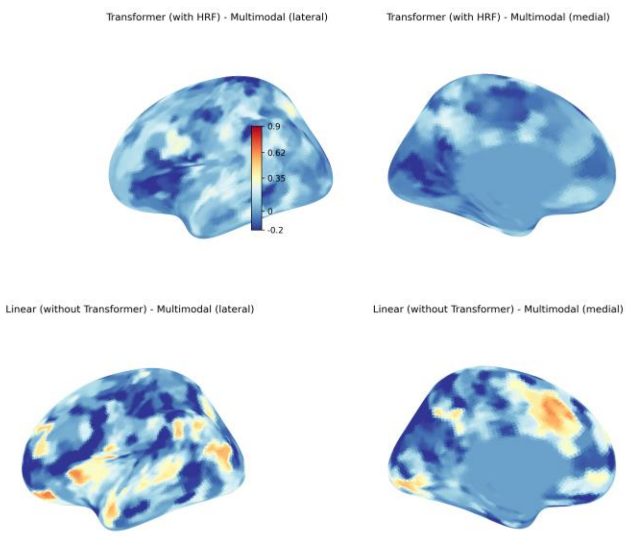

Fig. 3. Correlation maps for sub-01 test (multimodal). The correlations are predominantly low (r<0.15 in ~80% of parcels), without clear concentration in specific functional networks. This contrasts with the validation maps, which showed elevated correlations in visual and temporal regions.

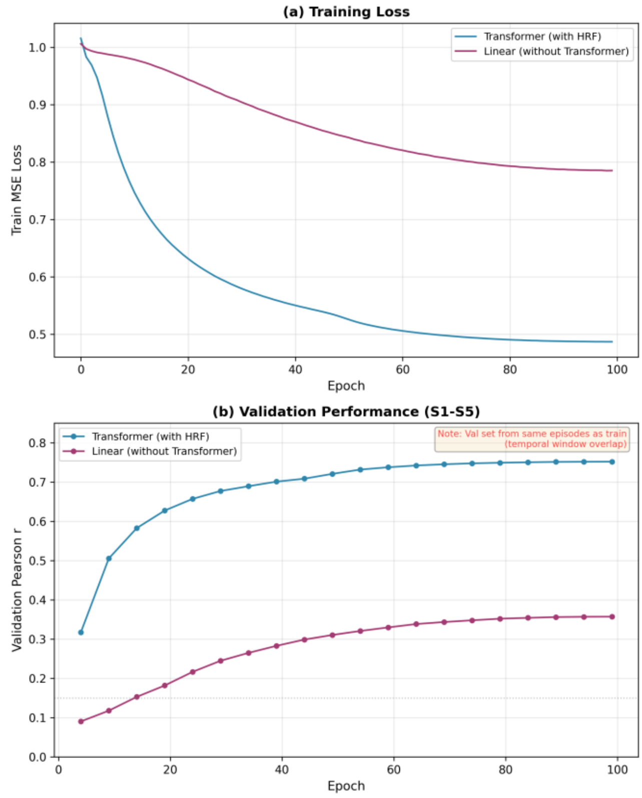

Fig. 4. Training curves (multimodal, sub-01). The Transformer reduces the MSE loss from 1.00 to 0.49 and reaches a val Pearson of 0.75 (epoch 100). Note: these curves reflect the validation set with leakage. The dotted lines indicate the actual performance on the test set (~0.07).

IV. Discussion

The central finding of this work is that embeddings extracted from Gemma 4 E2B-it, an instruction-tuned multimodal LLM [10], capture high-level narrative information (specific dialogs, scene context) but lack the low-level sensory representations that the brain uses to process natural stimuli [1], [14]. There are three key observations; the linear Ridge model achieves comparable or superior performance to a 14M parameter Transformer. This indicates that the relevant information is present in the features in a linearly decodable way, but there are no complex temporal structures that a Transformer can exploit [15].

Removing audio and image from the prompt reduces test performance by less than 0.01 Pearson. This suggests that Gemma 4 fuses all three modalities into a unified "story" representation [13], [16], where text dominates, and visual/auditory inputs provide redundant or indistinguishable information. The difference of ~0.40 Pearson between the Transformer and Linear in validation disappears completely in test. In validation, the Transformer memorizes specifically temporally correlated associations, a known challenge in the Out-of-Distribution (OOD) generalization of fMRI data [5], [8], which do not exist in Friends season 6 (test set).

Our result suggests an important distinction between two types of features for brain encoding. There are low-level sensory features that capture edges, movements, pitch, and phonemes; for these cases, deep architectures specialized in each modality that maintain high temporal fidelity are ideal, such as those used by TriBE [3], [4]. Another type of features that we can classify are the high-level, narrative ones, which capture semantics and intentions. These do not generalize between episodes with different content and are precisely the type of abstract representations that LLMs like Gemma 4 naturally produce [2]. To obtain robust brain encoding, a combination of both is required: sensory features for early cortical regions and narrative features for association regions [14], [17].

An important limitation of this work is the feature extraction strategy. Gemma 4 contains specialized sensory encoders (150M parameter ViT for vision, Conformer for audio) that capture low-level representations before projection into the language space [14]. However, our hooks were applied to the deep layers of the LLM (20-35), where visual and auditory information had already been projected and fused into the semantic language space.

V. Conclusions

Gemma 4 demonstrates that generalist multimodal LLMs produce linearly decodable embeddings for brain encoding (~0.10 Pearson in strict hold-out), but these high-level narrative embeddings do not benefit from complex temporal architectures, nor do they generalize to unseen content. This work establishes a realistic upper bound for brain encoding with generalist models and highlights the need for low-level sensory features to achieve the robust generalization observed in specialized systems such as TriBE.

Future work should address the critical bottleneck identified in this study, the feature extraction strategy. Gemma 4 contains specialized sensory encoders that will help to achieve robust generalization, future approaches should extract pre-projection embeddings from the ViT and Conformer, keeping visual and auditory representations separate until late fusion at the brain encoding head, like TriBE's architecture. This would allow the decoder to learn modality-specific weights, assigning higher importance to visual features for occipital parcels and auditory features for temporal areas. Additionally, evaluation should move from cross-episode hold-out to cross-subject hold-out, where the model is trained on one subject and tested on another, forcing the model to learn only stimulus-driven components that generalize across brains rather than subject-specific.

VI. References

[1] N. Kriegeskorte, "Deep Neural Networks: A New Framework for Modeling Biological Vision and Brain Information Processing," *Annu. Rev. Vis. Sci.*, vol. 1, pp. 417-446, 2015.

[2] J. Tang, M. Du, V. A. Vo, V. Lal, and A. G. Huth, "Brain encoding models based on multimodal transformers can transfer across language and vision," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[3] S. d'Ascoli, J. Rapin, Y. Benchetrit, H. Banville, and J.-R. King, "TRIBE: TRImodal Brain Encoder for whole-brain fMRI response prediction," *arXiv preprint*, 2026.

[4] S. d'Ascoli *et al.*, "A foundation model of vision, audition, and language for in-silico neuroscience," *Technical Report*, 25 Mar. 2026.

[5] P. S. Scotti and M. Tripathy, "Insights from the Algonauts 2025 Winners," *Medical AI Research Center (MedARC)*, 14 Aug. 2025.

[6] Courtois NeuroMod Team, "MRI," *Courtois NeuroMod 2020-beta documentation*, 2022. Online.

[7] Courtois NeuroMod Team, "Courtois NeuroMod," *Courtois NeuroMod 2020-beta documentation*, 2022. Online.

[8] A. T. Gifford *et al.*, "The Algonauts Project 2025 Challenge: How the Human Brain Makes Sense of Multimodal Movies," *arXiv preprint*, 2024.

[9] A. Schaefer *et al.*, "Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI," *Cerebral Cortex*, vol. 28, no. 9, pp. 3095-3114, 2018.

[10] Google DeepMind, "Gemma 4 model card," *Google AI for Developers*, 17 abr. 2026. Online.

[11] M. Grootendorst, "A Visual Guide to Gemma 4," *Exploring Language Models*, 3 abr. 2026. Online.

[12] D. A. Handwerker, J. Gonzalez-Castillo, M. D'Esposito, y P. A. Bandettini, "The continuing challenge of understanding and modeling hemodynamic variation in fMRI," *Neuroimage*, vol. 62, no. 2, pp. 1017-1023, 2012.

[13] S. R. Oota, K. Pahwa, M. Marreddy, M. Singh, M. Gupta, and B. S. Raju, "Multi-Modal Brain Encoding Models for Multi-Modal Stimuli," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[14] Y. Hu y Y. Mohsenzadeh, "Neural processing of naturalistic audiovisual events in space and time," *Communications Biology*, 2025.

[15] A. Vaswani *et al.*, "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NIPS)*, 2017.

[16] C. Camastra, A. Pelagi, A. Quattrone, y A. Sarica, "Benchmarking Multimodal Deep Fusion Strategies for Heterogeneous Neuroimaging and Cognitive Data Using a Controlled Sex Classification Task," *Brain Sciences*, vol. 16, no. 405, 2026.

[17] Y. Benchetrit, H. Banville, J.-R. King, "Brain decoding: toward real-time reconstruction of visual perception," *FAIR at Meta / PSL University*.